

CONFIDENTIAL



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER I 202x/202x

SUBJECT CODE : **SCSR1013/SECR 1013**

SUBJECT NAME : **DIGITAL LOGIC**

SECTION : **1/2/3/4 (SOFTWARE ENGINEERING)**
1/2/3/4/5/6/7/8/9 (COMPUTER SCIENCE)

TIME : **3 HOURS**

DATE/DAY :

VENUES :

INSTRUCTIONS:

1. There are SIX (6) structure questions.
2. Answer ALL questions in Answer Booklet.
3. Answer Question 2(b), 2c(ii), 3(ii), 6(b) using template provided in Appendix.
3. A candidate who is suspected of cheating in examinations is liable to disciplinary action including (but not limited to) suspension or expulsion from the University. All materials and or devices which are found in violation of any examination rules and regulation will be confiscated.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **THIRTEEN (13)** printed pages excluding this page.

QUESTION 1 (20 MARKS)

- a) Using a 4-bit parallel adder, calculate the sum of the binary number 1101 and 0111.
- Draw and label the block diagram of the 4-bit parallel adder. In the diagram, label each bit value and position clearly.
(2 marks)
 - Determine the carry in, sum and the output carry. Label each clearly in the same block diagram. The value for C_{in} (LSB) is LOW.
(2 marks)
- b) For error detection, parity checker is used to determine whether the number of '1' is ODD or EVEN. If sender transmit binary data 1010 1101 to the receiver. Draw the parity checker circuit using XOR gate. Label the circuit with binary values to determine the parity bit either ODD or EVEN.
(4 marks)
- c) Refer 4-bit comparator in Figure 1. Compare the input binary values between 0101 (**A**) and 1010 (**B**) using a 4-bit comparator.

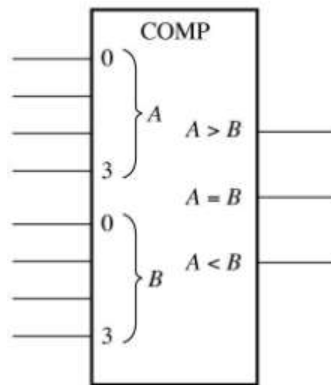
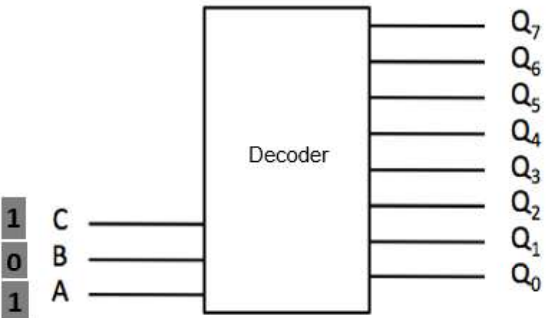
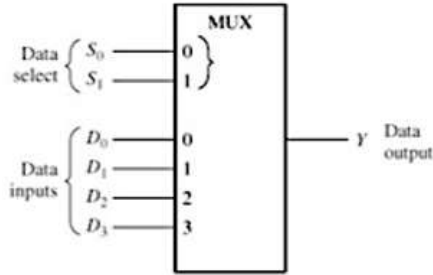
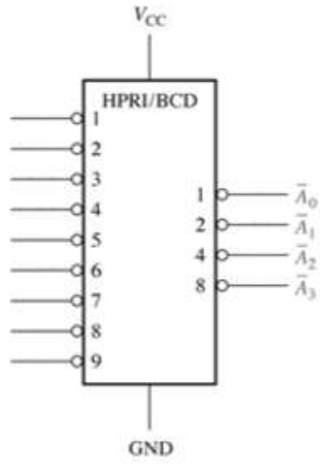
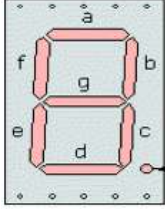
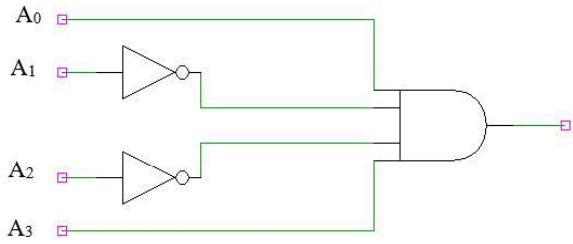


Figure 1: 4-bit comparator

- Re-draw Figure 1 and label the MSB and LSB position for both A and B.
(2 marks)
- Explain in details at which bit for A and B that comparator has to stop comparing?
(3 marks)
- What is the comparator output result?
(1 mark)

d) Identify the statement, either TRUE (T) or FALSE(F). Answer in answer booklet.

No	Statement
i.	A MUX is used to distribute a single input signal to multiple outputs
ii.	Refer to the following logic symbol. If binary value of ABC = 101, will data output Q₅ be activated (HIGH). 
iii.	Refer to the following logic symbol. If binary value of S₁S₀ = 01, data output Y will be selected from data input D₁. 
iv.	Refer to the following logic symbol of an HPRI/BCD encoder. If multiple inputs are active, such as when Q ₃ , Q ₅ , and Q ₇ are LOW, encoder produces BCD output that corresponds to the higher decimal value, the output of encoder will choose Q ₇ at the input and ignores the Q ₃ , and Q ₅ .

	
v.	Refer to the following logic symbol of a 7-segment display decoder. If the binary BCD inputs are 0001, segment <i>f</i> will light up (ON). 
vi.	Refer to the following logic circuit diagram. The circuit will produce an active HIGH output if the binary input for $A_3A_2A_1A_0$ is 1011. 

(6 marks)

QUESTION 2 (20 MARKS)

- (a) Identify the statement, either TRUE (T) or FALSE(F) regarding latches and flip-flops. Answer in answer booklet.

No	Statement
i.	A gated (EN) S-R flip-flop enters the SET state when S is HIGH, R is LOW, and EN is HIGH.
ii.	The S-R flip-flop has three output valid states.
iii.	The output of a J-K flip-flop does not change when both inputs are HIGH.
iv.	The $\overline{PRE}$ and $\overline{CLR}$ are known as asynchronous input.

(4 marks)

- (b) Draw the circuit for a T flip-flop using a J-K flip-flop and complete the output of the T flip-flop truth table as shown in Table 1. Answer in appendix.

(4 marks)

Table 1: T flip-flop truth table

INPUT				Output
CLK	$\overline{PRE}$	$\overline{CLR}$	T	Q
↑	0	0	1	
↑	0	1	1	
↑	1	0	0	
↑	1	1	0	
↑	1	1	1	
↑	1	1	1	
↑	1	1	0	

- (c) Consider a circuit diagram that includes a J-K flip-flop (positive triggered), a D flip-flop (positive triggered), and OR gate logic, as illustrated in Figure 2. The synchronous inputs

to the circuit are J and K, while the asynchronous inputs are $\overline{PRE}$ and $\overline{CLR}$. Both flip-flops are triggered by the same clock signal, CLK.

- i. Write equation for input D and output Z.
- ii. Given the asynchronous and synchronous inputs shown in Table 2, determine the value of input D and output flip-flops Q1, Q2, and output Z. Answer in appendix.

(12 marks)

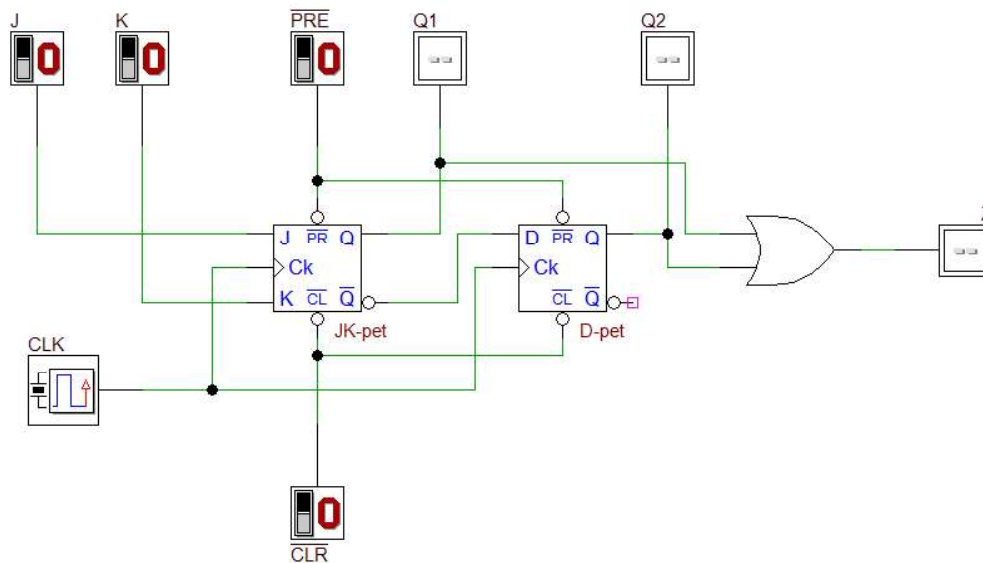


Figure 2: Asynchronous J-K flip-flop and D flip-flop

Table 2: Asynchronous and synchronous inputs and outputs

Input Flip-Flop						Output	State	Output	Output
CLK	$\overline{PRE}$	$\overline{CLR}$	J	K	D	Q1	Q1	Q2	Z
-	0	1	0	0					
-	1	0	1	1					
↑	1	1	0	0					
↑	1	1	0	1					
↑	1	1	1	0					
↑	1	1	1	1					

QUESTION 3 (12 MARKS)

Design a MODULUS-5 count-up asynchronous counter using positive edge-triggered JK flip-flops with asynchronous input $\overline{PRE}$ and $\overline{CLR}$.

- Draw the logic circuit. (6 marks)
- Draw the waveform outputs for 6 clock cycles if $\overline{PRE}$ and $\overline{CLR}$ are HIGH. Answer in appendix. (6 marks)

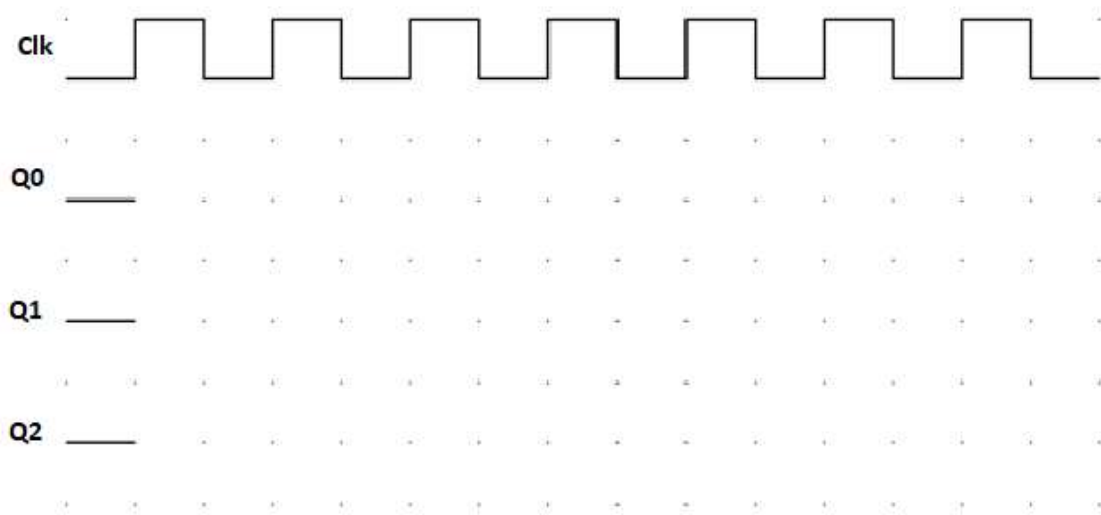


Figure 3: Waveform output

QUESTION 4 (16 MARKS)

a) Refer to state diagram in Figure 4. Design a synchronous counter using D flip-flops with input X and output Z:

- Construct a complete next state and D flip-flop transition table. Use header table as shown in Table 3. (7 marks)
- Draw K-maps to get minimum Boolean expression using SOP. (4 marks)
- Draw the complete circuit diagram design. (3 marks)

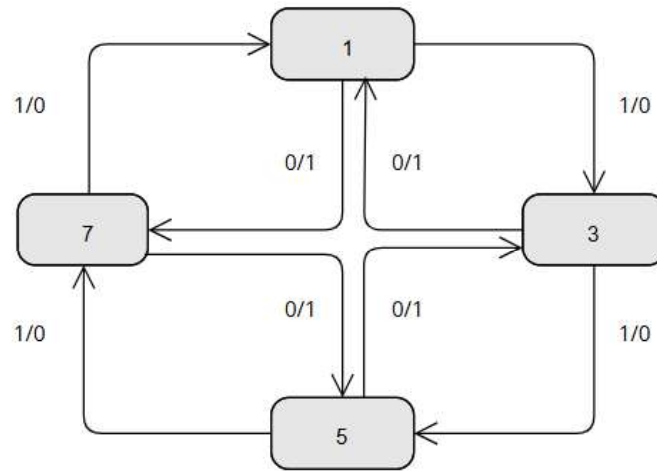


Figure 4: State Diagram

Table 3: Next state and flip flop transition table

Input	Present State	Next State	Input Flip Flop	Output
X				Z

b) Calculate the output frequency in MHz produced by the cascaded counter of MOD-4 and MOD-4 if an initial frequency input to the cascaded counter is given as 8GHz?

(2 marks)

QUESTION 5 (16 MARKS)

Refer circuit in Figure 5.

- a) Derive Boolean expression for the input flip-flop J1, K1, J0, K0 and output Z.

(5 marks)

- b) Produce the next state and flip flop transition table. Use the following Table 4 header.

(7 marks)

Table 4: Next state and flip flop transition table

Input	Present State		Next State		Input Flip Flop				Output
X	Q1	Q0	Q1(+)	Q0(+)	J1	K1	J0	K0	Z

- c) Draw a complete state diagram for the circuit shown.

(4 marks)

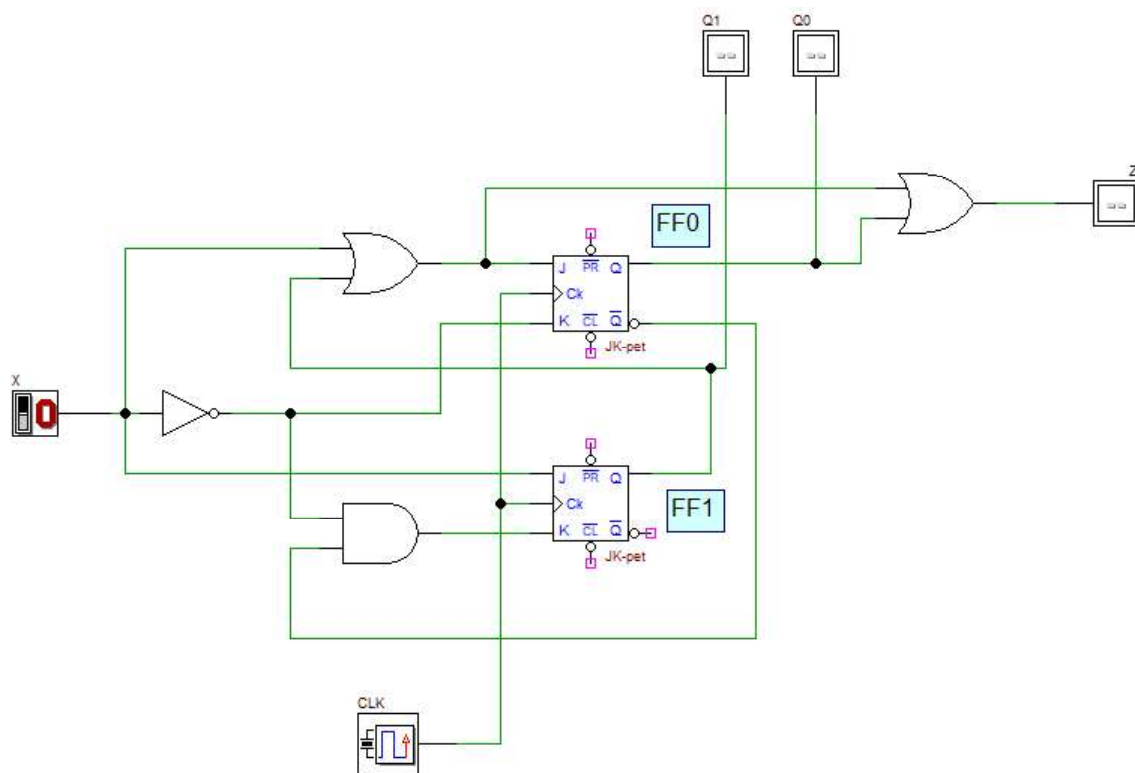


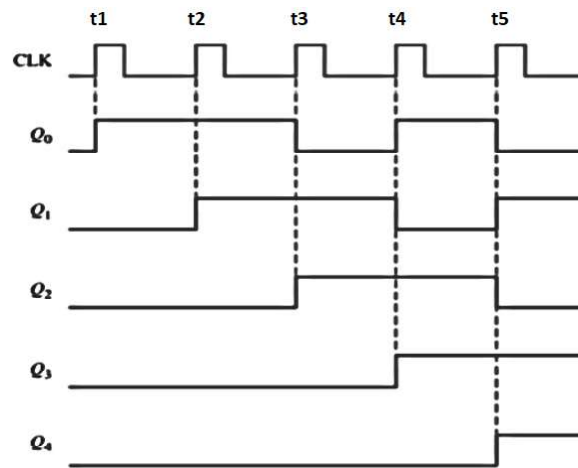
Figure 5: Synchronous counter

QUESTION 6 (16 MARKS)

a) Identify the statement, either TRUE (T) or FALSE(F). Answer in Answer Booklet.

No	Statement
i.	Refer to the following logic symbol. The number of clock required to shift in and out the binary data from shift register is equal to 8 clock.
ii.	Refer to the following shift register logic symbol. If we set value of $\overline{\text{SHIFT/LOAD}}$ to LOW, data output of Q3 become HIGH when clock is triggered.
iii.	Refer to the following logic symbol. The data is ready for output after fourth clock cycle.
iv.	All type of shift registers (SISO, SIPO, PISO, PIPO) operate based on ripple clock.

- v. Refer to the following timing diagram. The valid data out for SISO shift register (output Q0 to Q4) occurs at t4 clock triggered.



(5 marks)

b) A shift register counter is a type of digital sequential circuit that combines the principles of a shift register and a counter to generate a sequence of states or outputs.

- i. Draw a 3-bit Ring Counter with negative edge triggered and a positive edge triggered Johnson Counter circuit diagram using D flip flops. (6 marks)
- ii. Complete the output of 3-bit Ring Counter and 3-bit Johnson Counter as shown in Table 5 and Table 6. Answer in appendix. (4 marks)

Table 5: 3-bit Ring Counter

Clock	Q0	Q1	Q2
initial	0	1	0
↓			
↓			
↓			
↓			

Table 6: 3-bit Johnson Counter

Clock	Q0	Q1	Q2
initial	0	1	0
↑			
↑			
↑			
↑			

iii. Calculate number of valid states produced by a 3-bit Ring and 3-bit Johnson counters. (1 mark)

----- END of QUESTIONS -----

APPENDIX:

Name : _____

Matric No : _____

Section : _____

Answer Question 2(b)

Table 1: T flip-flop truth table

INPUT				Output
CLK	$\overline{PRE}$	$\overline{CLR}$	T	Q
↑	0	0	1	
↑	0	1	1	
↑	1	0	0	
↑	1	1	0	
↑	1	1	1	
↑	1	1	1	
↑	1	1	0	

Answer Question 2c (ii)

Table 2: Asynchronous and synchronous inputs and outputs

Input Flip-Flop						Output	State	Output	Output
CLK	$\overline{PRE}$	$\overline{CLR}$	J	K	D	Q1	Q1	Q2	Z
-	0	1	0	0					
-	1	0	1	1					
↑	1	1	0	0					
↑	1	1	0	1					
↑	1	1	1	0					
↑	1	1	1	1					

Answer Question 3(ii)

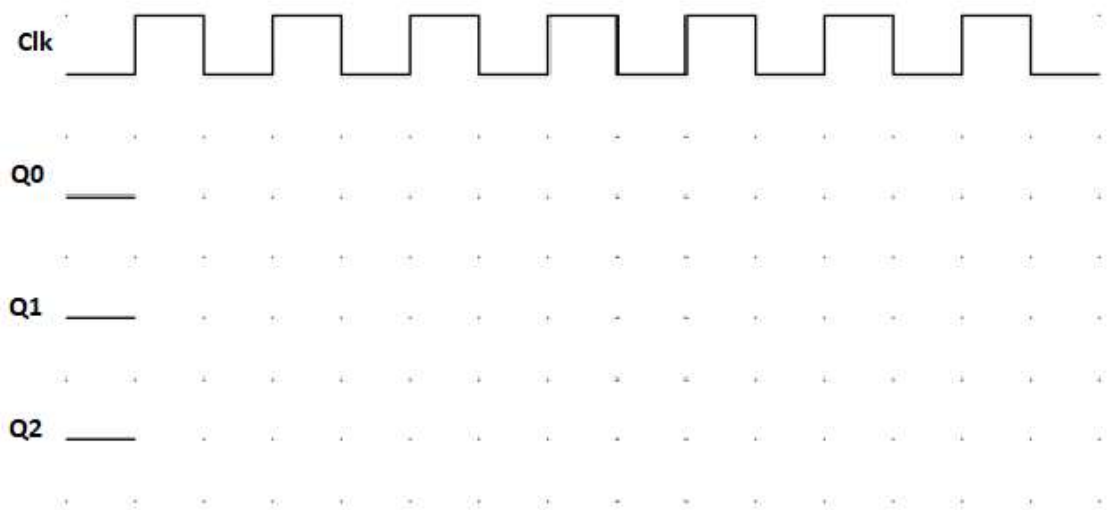


Figure 3: Waveform output

Answer Question 6 (b)

Table 5: 3-bit Ring Counter

Clock	Q0	Q1	Q2
initial	0	1	0
↓			
↓			
↓			
↓			

Table 6: 3-bit Johnson Counter

Clock	Q0	Q1	Q2
initial	0	1	0
↑			
↑			
↑			
↑			